

PREDICTING REVENUE FROM HOTELS ANALYSIS REPORT

STAT 101 Midterm

PREPARED BY:

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Executive Summary

The Australian Tourism Bureau desires to predict revenue by hotel chains. Additionally, they wish to discontinue collection of the sources of data that are least useful for predicting revenue. A standard linear least squares regression model for predicting revenue was created using the following parameters: Bed spaces, Rooms, Guest nights, Room nights occupied, and Bed occupancy rate. Persons Employed and Average length of stay were not useful for predicting revenue. Unless those parameters are useful in another way collection of those data is no longer necessary.

Predictive Model Prediction Expression for Revenue:

-2180081.234
+ 15.855081445 • Bed spaces
+ -21.81591478 • Rooms
+ -312.3290478 • Guest nights ('000)
+ 546.338636 • Room nights occupied ('000)
+ 28489.538427 • Bed occupancy rate (%)

Details of Analysis and Recommendations

Data Preparation:

RECODING:

For column "Room occupancy rate (%)."

The value of 620 changed to 62.0 (row 1).

The value of 700 changed to 70.0 (row 28).

For column "Average length of stay (days)."

Two values of 22 changed to 2.2 (row 35 & row 52).

For column "Persons Employed."

The value of 7088 changed to 70880 (row 27).

ADDRESSING MISSING VALUES:

For column "Room occupancy rate (%)."

Two missing values changed to 66.575675676 (row 22 & row 48).

Action was performed Using JMP Automated Data Imputation.

CHECKING OUTLIERS:

For column "Average length of stay (days)."

The value of 2.5 was identified as a significant but reasonable outlier (row 11).

Exploratory Data Analysis:

The data set included the following features:

- Bed Spaces
- Rooms
- Persons Employed
- Guest arrivals
- Guest nights
- Room nights occupied
- Revenue
- Room occupancy rate
- Bed occupancy rate
- Average Length of Stay

Strong Correlations:

- Bed Spaces & Rooms (.9804)
- Guest arrivals & Guest nights (.9609)
- Guest arrivals & Room nights occupied (.9861)
- Guest arrivals & Revenue (.9433)
- Room nights occupied & Guest nights (.9811)
- Room nights occupied & Revenue (.9555)

Although “Guest arrivals” and “Guest nights” are the most strongly correlated parameters with revenue (correlation coefficients $> .9100$), they are also strongly correlated with each other (.9609). Adding both parameters to a model would create additional noise without adding unique information. As a result, only “Guest nights” was used in the model to predict revenue.

Results of Analysis:

CREATING THE MODEL:

A stepwise personality on JMP was used to identify the best predictors of revenue:

Step	Parameter	Action	"Sig Prob"	Seq SS	RSquare	Cp	p	AICc	BIC	RSquare Validation	
1	Room nights occupied ('000)	Entered	0.0000	1.05e+12	0.9170	29.988	2	1007.3	1011.79	0.9356	○
2	Guest nights ('000)	Entered	0.0001	3.31e+10	0.9458	8.7756	3	992.321	998.064	0.9324	○
3	Bed spaces	Entered	0.1499	3.443e+9	0.9487	8.359	4	992.596	999.449	0.9446	○
4	Rooms	Entered	0.0423	6.46e+9	0.9544	5.8249	5	990.591	998.402	0.9454	○
5	Bed occupancy rate (%)	Entered	0.0495	5.548e+9	0.9592	3.9308	6	988.932	997.533	0.9619	○
6	Imputed_Room occupancy rate (%)	Entered	0.3034	1.46e+9	0.9605	4.906	7	990.742	999.951	0.9299	○
7	Guest arrivals ('000)	Entered	0.4563	7.695e+8	0.9611	6.366	8	993.349	1002.96	0.9131	○
8	Average length of stay (days)	Entered	0.5730	4.483e+8	0.9615	8.0513	9	996.462	1006.26	0.9222	○
9	Persons Employed	Entered	0.8223	73096482	0.9616	10	10	1000.16	1009.91	0.9228	○
10	Best	Specific	.	.	0.9592	3.9308	6	988.932	997.533	0.9619	●

A model was created using a standard linear least squares regression using the five best parameters.

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CHECKING THE MODEL:

The model fits the data well with an R Square value of .959191.

The model is not overfit as the R Square values for the Training and Validation sets are similar.

Training Set R Square = .9592

Validation Set R Square = .9619

The Model is trustworthy as the Prob>|t| values for each component are less than .05.

Intercept = .0006

Bed spaces = .0020

Rooms = .0166

Guest nights ('000) = .0005

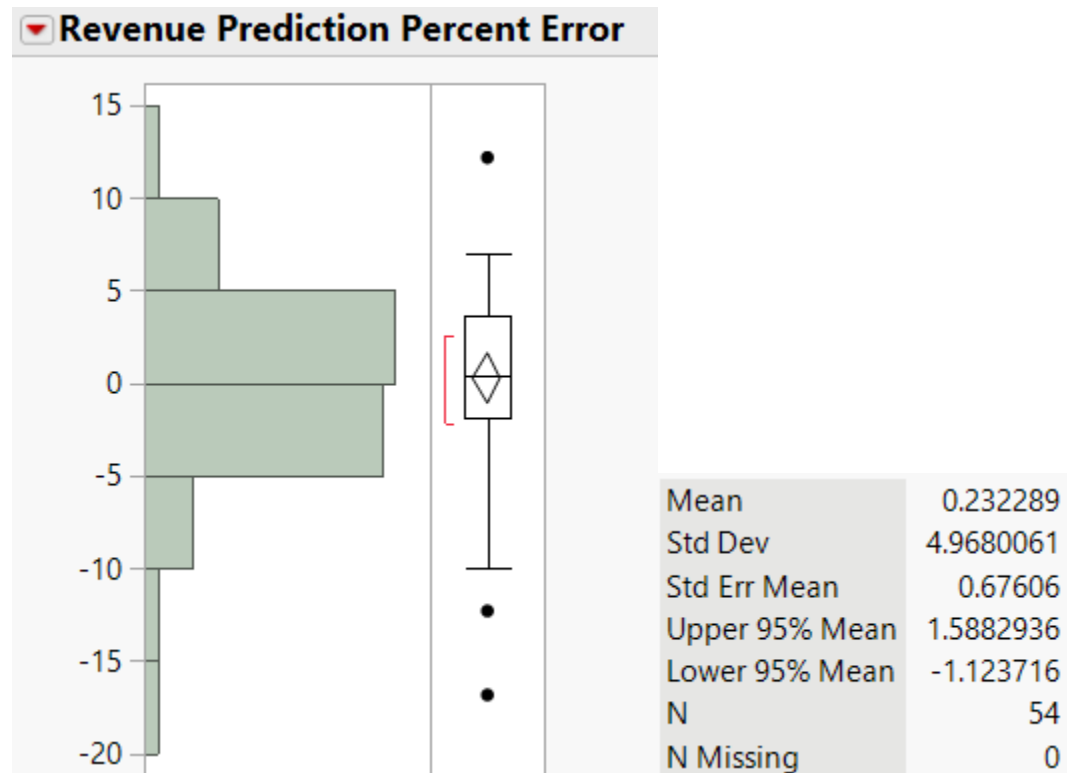
Room nights occupied ('000) <.001

Bed occupancy rate (%) = .0495

MODEL PERCENT ERROR DISTRIBUTION:

Percent error calculation:

$$((\text{Predicted Revenue}/\text{Actual Revenue})-1) * 100$$



Outlier Analysis for Prediction Percent Error:

The |percent error| was greatest (12-17%) for small hotel chains in the data set. (rows 1, 2, & 10). This may suggest the model is less effective at predicting revenue for small (less than 75000 rooms) hotel chains.

OTHER PARAMETER CONSIDERATIONS:

The stepwise personality in JMP identified “Room occupancy rate (%)” as the next best parameter for predicting revenue, so a different model that included Room occupancy rate (%) as well as all other parameters was tested. However, while the R Square value increased from .959191 to .960461 the trustworthiness of the “Room occupancy rate (%)” parameter was insufficient ($\text{Prob} > |t| = .3034$). A $\text{Prob} > |t| = .05$ cut off was used for accepting the trustworthiness of parameters. Due to this, the final model does not include the “Room occupancy rate (%)” parameter.

EXAMPLE PREDICTIONS:

Row 1 Predicted Revenue: \$439680.23546.

Row 1 Actual Revenue: \$501266.90.

Percent Error: -12.28620213%.

Percent error calculation: $((\text{Predicted Revenue}/\text{Actual Revenue})-1) * 100$.

Prediction for new row not in initial data set:

Bed spaces	Rooms	Persons Employed	Guest arrivals ('000)	Guest nights ('000)	Room nights occupied ('000)	Revenue (\$'000)	Room occupancy rate (%)	Bed occupancy rate (%)	Average length of stay (days)
170000	68265	71886	2675.7	6000	3789.5	????	62	37.7	2.3

Predicted Revenue for new row: \$296450.76288.

RECOMMENDATIONS:

The usefulness of each column was assessed based on JMPs stepwise personality. The helpfulness of the columns in the data set in predicting revenue, from most helpful to least helpful, were Room nights occupied, Guest nights, Bed spaces, Rooms, Bed occupancy rate, Room occupancy rate, Guest arrivals, Average length of stay, and Persons Employed.

Correlation coefficients between each column and revenue were also examined. "Persons Employed" and "Average length of stay (days)" were found to be very weakly correlated with revenue ($| \text{correlation coefficients} | < .5$).

Both the stepwise personality and correlation coefficients suggest that "Persons Employed" and "Average length of stay (days)" are not useful for predicting revenue. Unless they are useful for predicting other factors, there is no need to continue collecting data for those parameters.

Stepwise Analysis:

Correlation Coefficients for Revenue:

Step	Parameter	Revenue (\$'000)
1	Room nights occupied ('000)	1.0000
2	Guest nights ('000)	0.8850
3	Bed spaces	0.8830
4	Rooms	-0.4137
5	Bed occupancy rate (%)	0.9500
6	Imputed_Room occupancy rate (%)	0.9151
7	Guest arrivals ('000)	0.9607
8	Average length of stay (days)	0.8118
9	Persons Employed	0.7483
		-0.4894

Conclusion:

The following model can be used to predict revenue. The model was created using a standard linear least squares regression with five parameters selected through a stepwise personality. The model fits the data well, performed well with a validation test, and has been deemed trustworthy. However, the model did have a large percent error for 3 of the 10 smallest hotel chains in the data set. When considering variables for the model "Persons Employed" and "Average length of stay" were not useful predictors. This model allows for a prediction, but the hotels used on the model should be similar in size to the hotels in the data set (Between 68000 and 86000 rooms).

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